

Time-regularised Lagrange multipliers in MGD: audit, corrections and the *moment* energy

Chiara Zelco (prepared with Claude Code)

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Provenance tags. [\[verified\]](#): derived in this note and/or checked by a script in `notes/guth_reg_audit_0930/`, whose output is quoted. [\[reported, not re-run\]](#): from the 29 Sep notes or the 30 Sep review agent, not re-run. [\[derived, external review\]](#): stated by the 30 Sep external review (checked against the MGD PDF project copy, Guth et al. and the draft *ExtremeScattering-7*); not re-checked here, because those documents are not available to me. [\[hypothesis\]](#): not tested.

Sign conventions. Every formula states its sign. MGD sign: $p_\theta \propto e^{-\theta^\top \phi}$. CODE sign: $\theta^{\text{code}} = -\theta$, i.e. $p \propto e^{+\theta^{\text{code}} \cdot \phi}$ (what `theta_t`, `Theta_reg` and the saved systems hold).

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1 Summary

- The time term of Θ_{reg} used so far (**legacy**) is uninformative. It is built from the conditional time score τ of Guth et al., evaluated on walkers that are independent of the noise Z , so it

targets a constant (Section 5.1). Every past Θ_{reg} , including the bimodal REG figures (A8), was smoothed toward that target.

- New energy `moment` (now the default with `--reg.solver thomas`). Its time term is time-score matching on the walker law, whose normal equations are exact by the MGD constraint alone (A4): $\hat{\Sigma}_w \dot{\Theta} = -\hat{m}$ (MGD sign). It uses no τ , costs nothing extra per step, and can be rebuilt offline for old runs.
- Implemented B1–B8 (Section 4). Unit and end-to-end tests pass. The legacy path is bit-identical to 4baf4c2 (Section 6.1).
- Validation in 1-D against a Fokker–Planck ground truth (C1, C2) and a check of the per-step noise model (B4): Section 6. C3 (existing turbulence systems) needs Jean Zay; the commands are in Section 6.4.

2 Corrections (30 Sep review)

A1. Algorithm 1 corrector sign: closed.

The PDF has the same line as the HTML, $x_{k+1} = y_k + h\hat{\theta}^\top \nabla \phi(y_k)$. It is a typo in the paper, wrong in both the sign and the missing σ^2 [derived, external review]. The correct corrector is MGD eq. (19), $X_{t+h} = Y - h\sigma^2 \hat{\theta}^\top \nabla \phi(Y)$ (MGD sign), with $\hat{\theta}$ from eq. (21) [derived, external review]. The code follows eq. (19) [verified] (`x_k_plus_one = y_k + theta_raw.grad phi`, $\theta_{\text{raw}} = h\sigma^2 \theta^{\text{code}} = -h\sigma^2 \theta$). All notes now cite eqs. (19)/(21), not Algorithm 1.

A2. Numbering.

In the project copy: $D_{\text{KL}}(p_1^\sigma \| p_*) \leq C\sigma^{-2}$ is Conjecture 4.1 (eq. 25); $\frac{d}{dt}H \geq \theta^\top \dot{m}$ is Proposition 4.3 (eqs. 28–29); the entropy bound is Conjecture 4.4 (eq. 32) [derived, external review]. The arXiv HTML version I read numbers them Conj. 4.3 / Prop. 4.4 / Conj. 4.5; this note uses the project-copy numbers. “Nothing is claimed for intermediate t ” was too strong: the formal $\sigma \rightarrow \infty$ limit (eqs. 22–24) holds for all t , and only the conjecture is stated at $t = 1$ [derived, external review].

A3. The premise of the old §3.1 was wrong; its conclusion stands.

The MGD constraint is $\mathbb{E}[\phi(X_t)] = m_t$ for the walker law p_t^σ (eqs. 12, 17), not $\mathbb{E}_{p_{\theta_t}}[\phi] = m_t$ [derived, external review]. MGD’s θ_t solves $G_t \theta_t = \mathbb{E}[\Delta \phi(X_t)]$ (eq. 15, MGD sign): it is the Fisher (score-matching) projection of p_t^σ onto the family, with the gap given by eq. (31) [derived, external review]. The identity $\Sigma_\theta \dot{\theta} = -\dot{m}$ holds for the max-entropy multiplier $\theta^{\text{ME}}(m_t)$, which equals θ_t only formally as $\sigma \rightarrow \infty$. Re-derived [verified]. Numerically, in 1-D (bimodal, Fokker–Planck ground truth), $\|\theta_t - \theta^{\text{ME}}\|_G / \|\theta^{\text{ME}}\|_G = 1.35 \cdot 10^{-3}$ at $\sigma = 10$ and $3.6 \cdot 10^{-2}$ at $\sigma = 2$ [verified]. The sign and centring conclusion (legacy targets $K \dot{\Theta} = -\mathbb{E}[\phi \tau]$ with the wrong sign and uncentred) stands.

A4. The proposed P2 is exact; it is the draft’s eq. (6).

Take time-score matching on p_t^σ with model time score $-\dot{\Theta}^\top \phi + c_t$, where c_t is free (MGD sign). Its normal equations are

$$\Sigma_w \dot{\Theta} = -\text{Cov}_w(\phi, \partial_t \log p_t^\sigma) \quad \text{and} \quad \text{Cov}_w(\phi, \partial_t \log p_t^\sigma) = \frac{d}{dt} \mathbb{E}_w[\phi] = \dot{m}_t$$

by the constraint alone. Re-derived [verified] ($\mathbb{E}_w[\partial_t \log p_t^\sigma] = 0$ and $\frac{d}{dt} \mathbb{E}_w \phi = \mathbb{E}_w[\phi \partial_t \log p_t^\sigma]$). It is the draft’s eq. (6), derived in its Appendix [derived, external review]. Consequences:

- “unbiased when $\Sigma_w = \Sigma_\theta$ ” and “not a TSM on a single law” are deleted;
- the old open question 1 is answered: $C = \hat{\Sigma}_w$;
- empirically the identity holds to $O(h)$, because the corrector enforces the moment condition on walker averages [derived, external review].

What remains is a *projection mismatch*. The data term targets the Fisher projection of $p_t^\sigma(\theta_t)$; the time term targets its time-score projection. They coincide only when p_t^σ is in the family (formally $\sigma \rightarrow \infty$). The earlier wording calling the dynamics model “nearly exact” is withdrawn. The τ -based modes were a mis-implementation of the draft’s eq. (4) [derived, external review].

A5. The old C5 was not a test of MGD’s θ_t .

In `verify_bias.py`, θ is obtained by moment matching, i.e. it is θ^{ME} . The 10^{-10} row checks the quadrature and the finite differences only [verified]. Restated conclusion: the interpolant-side TSM projects the time score of the *interpolant* law p_t , while the data term projects the score of p_t^σ . It mixes two laws. The recommendation to drop it stands.

A6. Guth invariance (old open question 4): resolved; λ stays free.

(a) Guth’s loss written in Cos time is invariant. Data rescaling $X \rightarrow cX$ gives $Y' = cY_\beta$ with $\tan \beta = c \tan \alpha$, which is Guth’s rescaling. I checked this algebra [verified]: $Y' = cX + \cot \alpha Z = c(X + \cot \beta Z)$. The improper $p(s)$ is not a new problem: $w_D = 2\dot{\alpha} \cos \alpha / (d \sin \alpha) \approx 2/(dt)$ as $t \rightarrow 0$ [verified], the same log divergence as $\int ds/s$, with t_1 in the role of s_{max} [derived, external review]. (b) Our estimator is not invariant at finite σ , whatever the weights. The rescaling acts as a time reparametrisation plus a time-dependent dilation, which turns constant- σ MGD into MGD with a time-dependent effective volatility. Invariance returns only formally as $\sigma \rightarrow \infty$, and only for homogeneous ϕ_k [derived, external review]. So invariance does not justify a fixed trade-off. The w_D , w_T formulas (with the $\sin^2 \alpha$ fix) were re-derived by the review and are correct [derived, external review]; they were already checked here [verified].

A7. Entropy and $\log \mathcal{Z}$: keep the per-step θ .

Replacing θ by Θ_{reg} inside H^* (MGD eq. 30) or the draft’s $\log \mathcal{Z}$ bound (eq. 8) loses the lower-bound guarantee. In the MGD sign, $\frac{d}{dt}H - \Theta^\top \dot{m} = (\theta - \Theta)^\top \dot{m} + \sigma^2 \cdot$ (Fisher gap), and the first term has no sign [derived, external review]. The per-step noise averages out in these integrals anyway. With the B4 noise model, $\text{Var}(\sum_k h \hat{\theta}_k^\top \dot{m}) \approx \frac{2}{n\sigma^2} \int_0^1 \dot{m}^\top G^{-1} \dot{m} dt$, finite as $h \rightarrow 0$ (derivation [verified], given B4).

Current use in the code [verified] (`codes/utils_entropy.py` and the notebooks calling it): H^* always integrates the per-step `dH_t_bound`; Θ_{reg} enters only as the end point θ_1 in $\log \mathcal{Z} \geq H^* + \theta_1^{\text{code}} \cdot m_1$ (`log_Z_bound(..., 'Theta_reg')`; `jets/jets_KL_divergence.ipynb`; `turbulence/turb_KL_divergence.ipynb`; `theta_interpretation/`). By Gibbs’ inequality, $\log \mathcal{Z}_\vartheta \geq H(q) - \vartheta^\top \mathbb{E}_q \phi$ holds for *any* ϑ (MGD sign; $q = p_1^\sigma$, $H(q) \geq H^*$). So that end-point use is still a valid lower bound, but on $\log \mathcal{Z}$ at $\vartheta = \Theta_{\text{reg}}(1)$, a different model from the one at θ_1 [verified] (derivation). Recommendation: per-step θ for H^* and $\log \mathcal{Z}$; Θ_{reg} for $\theta(t)$ and θ_1 as a coupling/interpretation object.

Side finding [verified]: `compute_log_Z_sigma` in `jets/jets_KL_divergence.ipynb` computes $H - \theta_1^{\text{code}} \cdot m_1$, while `utils_entropy.log_Z_bound` (and the derivation, CODE sign) gives $H + \theta_1^{\text{code}} \cdot m_1$. The notebook is not edited.

A8. Draft Fig. 1 (“regularised MGD, 2 orders of magnitude”): flagged for re-run.

I cannot see the draft. The bimodal REG pipeline is the candidate: `bimodal_experiment/kl_bimodal.ipynb` (`KL_bimodal_paper_1.png`, `LAM='selected'`) and `analyse_lamsweep.ipynb`, on `results_bim_theta_beta`. It solves the legacy energy (`codes/lam_selection.py`: `uncentred G, f[:-1] -= lam/dt c`) [verified], with λ from the z^2 rule. Its walkers start at $x_0 = Z$ (`xt = x0.clone()`, `sde_routines_scalar_reg.py`) [verified], so the time term is the “walkers started at Z ” case of Section 5.1: unrelated to θ . If Fig. 1 comes from this pipeline, its gain came from smoothing toward an unrelated target [hypothesis]. The bimodal port is not changed yet (see Section 7).

3 Notation

MGD (Lempereur et al., arXiv:2602.17211; equation numbers of the project copy [\[derived, external review\]](#)). $x \in \mathbb{R}^d$; statistics $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^r$; $Z \sim \mathcal{N}(0, \text{Id})$, $X \sim p$; interpolant $I_t = \cos \alpha_t Z + \sin \alpha_t X$, **Cos** schedule $\alpha_t = \pi t/2$, $\dot{\alpha} = \pi/2$. Target moments $m_t = \mathbb{E}[\phi(I_t)]$. Walkers $X_t \sim p_t^\sigma$, constraint $\mathbb{E}[\phi(X_t)] = m_t$ (eqs. 12, 17). Gram $G_t = \mathbb{E}[\nabla \phi(X_t) \nabla \phi(X_t)^\top]$. MGD sign: $G_t \eta_t = \dot{m}_t$, $G_t \theta_t = \mathbb{E}[\Delta \phi(X_t)]$ (eq. 15); SDE $dX_t = (\eta_t - \sigma^2 \theta_t)^\top \nabla \phi dt + \sqrt{2\sigma} dW_t$, $X_0 = Z$; Fokker–Planck form eq. (22). Corrector eqs. (19), (21).

Two multipliers (MGD sign). θ_t : MGD’s, the Fisher projection of p_t^σ (eq. 15). $\theta^{\text{ME}}(m)$: the max-entropy multiplier, $\mathbb{E}_{p_{\theta^{\text{ME}}}}[\phi] = m$. They are equal formally as $\sigma \rightarrow \infty$.

Guth et al. (arXiv:2506.05310). Their noise variance “ t ” is written s here. $y = x + \mathcal{N}(0, s\text{Id})$, $p \propto e^{-U}$; $\ell = \mathbb{E}_s[\frac{s}{d}\ell_{\text{DSM}} + (\frac{s}{d})^2\ell_{\text{TSM}}]$ with $p(s) \propto 1/s$ (their eq. 8); ∂_s in ℓ_{TSM} at fixed y .

Other symbols. $K = \mathbb{E}[\phi\phi^\top]$, $\Sigma = \text{Cov}(\phi)$; subscripts w (walker empirical law), I (interpolant); hats are Monte-Carlo estimates. n walkers, SDE step $h = t_{k+1} - t_k$, nodes t_k of the regularised problem, Δt_k node spacing, h_k trapezoid weights. $\tau(x; Z) = -\dot{\alpha}[\tan \alpha(d - \|Z\|^2) + Z^\top X]$, $X = (x - \cos \alpha Z)/\sin \alpha$: minus the conditional time score of $p_t(x|X)$ (sign-free).

symbol (CODE sign unless noted)	saved key	definition at node t_{k+1}
$\hat{M}_k$	M	$\frac{1}{n} \sum_i \nabla \phi(y_k^i) \nabla \phi(y_k^i)^\top (= \hat{G}_k)$, float32
$\hat{b}_k$	b	$(\hat{m}_{t_{k+1}} - \phi(y_k))/(h\sigma^2)$, float32; MGD sign: $\hat{\beta}_k = -\hat{b}_k$
$\hat{\Sigma}_{w,k}$	Sigma (moment)	$\frac{1}{n} \sum_i (\phi^i - \hat{m}_w)(\phi^i - \hat{m}_w)^\top$, $\phi^i = \phi(x_{k+1}^i)$, float64 accumulation, stored float32
$\hat{m}_{w,k}$	m	$\frac{1}{n} \sum_i \phi(x_{k+1}^i)$ (moment), $\phi(y_k^i)$ (legacy), float64
$\hat{m}_k$	mdot (moment)	$\frac{1}{n} \sum_i \nabla \phi(I_{t_{k+1}}^i) \cdot \dot{I}_{t_{k+1}}^i$ (sign-free), float64
h_k , live	h , live	SDE step ending at the node; per-step mask of the corrector
$\hat{K}_{w,k}$, $\hat{c}_{w,k}$, $\bar{\tau}$	G , c , tau_mean (legacy)	$\frac{1}{n} \sum \phi\phi^\top$, $\frac{1}{n} \sum \phi(y_k^i) \tau(y_k^i; x_0^i)$ at y_k

Table 1: Per-node quantities of the saved systems (`_save_reg_system`; `meta['reg_mode']` says which set).

4 The estimators as implemented

4.1 Per-step $\hat{\theta}_k$ and its noise (B4)

CODE sign: $(\hat{M}_k + \varepsilon D_k) \hat{\theta}_k^{\text{code}} = \hat{b}_k$, $D_k = \text{diag } \hat{M}_k$, $\varepsilon = \text{regularization}$; potentials below `_live_floor` are set to 0. MGD sign: $(\hat{M}_k + \varepsilon D_k) \hat{\theta}_k = \hat{\beta}_k \approx \mathbb{E}[\Delta \phi]$ (eq. 21).

Claim (noise model, [\[derived, external review\]](#); derivation [\[verified\]](#); holds numerically at $\sigma = 2$, Table 5). $\text{Cov}(\hat{\theta}_k) \approx \frac{2}{nh\sigma^2} \hat{M}_k^{-1}$, *white in k* .

Derivation: given x_k , the Brownian increment makes $\bar{\phi}(y_k)$ fluctuate with covariance $\approx \frac{2h\sigma^2}{n} \hat{M}_k$. Dividing by $h\sigma^2$ gives $\text{Cov}(\hat{\beta}_k) \approx \frac{2}{nh\sigma^2} \hat{M}_k$, hence the claim. It is white because each corrector step resets the walker averages to $\hat{m}_{t_{k+1}}$, so no fluctuation carries over to the next step.

4.2 Regularised energy

CODE sign (what `_solve_regularised.thomas` minimises; weights from `reg.energy_weights`):

$$\mathcal{E}[\Theta] = \sum_k a_k \left(\frac{1}{2} \Theta_k^\top \hat{M}_k \Theta_k - \hat{b}_k^\top \Theta_k \right) + \sum_{k < N} \left(\frac{q_k}{2} \Delta \Theta_k^\top C_k \Delta \Theta_k + l_k \tilde{c}_k^\top \Delta \Theta_k \right). \quad (1)$$

mode	a_k	q_k, l_k	C_k (cell k)	$\tilde{c}_k$ (cell k)
legacy (reproduction)	1	$\lambda/\Delta t_k^2, -\lambda/\Delta t_k$	$\hat{K}_{w,k}$	$\hat{c}_{w,k}$
moment, uniform (default)	h_k	$\lambda/\Delta t_k, \lambda$	$\frac{1}{2}(\hat{\Sigma}_{w,k} + \hat{\Sigma}_{w,k+1})$	$-\frac{1}{2}(\hat{m}_k + \hat{m}_{k+1})$
moment, Guth schedule	$h_k w_D(t_k)$	$\lambda w_T/\Delta t_k, \lambda w_T$	same	same

Table 2: Energies. Guth schedule: $w_D = 2\dot{\alpha} \cos \alpha / (d \sin \alpha)$, $w_T(t_{k+1/2}) = \sin \alpha \cos \alpha / (2\dot{\alpha} d^2)$, $\lambda_{\text{eff}} = w_T/w_D = \sin^2(\pi t/2)/(\pi^2 d)$. It is a comparison option: the default weight is uniform in t because MGD eq. (31) weights the Fisher gap uniformly in t [derived, external review].

Moment mode, both signs. Per cell, the moment time term is $\lambda w_T \Delta t [\frac{1}{2} \dot{\Theta}_c^\top C \dot{\Theta}_c - \dot{m}^\top \dot{\Theta}_c]$ (CODE), minimised at $C \dot{\Theta}^{\text{code}} = \dot{m}$. With $\Theta = -\Theta^{\text{code}}$ this is $C \dot{\Theta} = -\dot{m}$ (MGD), the draft’s eq. (6). Checked by unit test T2: exact to $O(\Delta t^2)$ for every $\lambda \in [0, 10^4]$, and the flipped sign fails [verified].

Interpretation (B4), [derived, external review]; to be checked. With the quadrature weights $a_k = h_k$ and the noise model, the data term is $\frac{2}{n\sigma^2}$ times the Gaussian negative log-likelihood of the $\hat{\theta}_k$: $\sum_k h_k \frac{1}{2} (\Theta_k - \hat{\theta}_k)^\top \hat{M}_k (\Theta_k - \hat{\theta}_k) = \frac{2}{n\sigma^2} \sum_k \frac{nh_k \sigma^2}{4} (\dots)$ (up to a constant) [verified] (algebra). The time term is then a Gaussian prior on the residual $r = \hat{\Sigma}_w \dot{\Theta} + \hat{m}$ (MGD sign) with precision $\lambda n \sigma^2 / 2$ in the $\hat{\Sigma}_w^{-1}$ metric. Consequence used in the selection: $\chi^2(\lambda) = \sum_k \frac{nh_k \sigma^2}{2} (\hat{\theta}_k - \Theta_k)^\top \hat{M}_k (\hat{\theta}_k - \Theta_k) / N_{\text{live}}$ should be $\lesssim 1$ when λ removes only noise.

Projection mismatch (A4). In the moment mode the data term pulls toward θ_t (Fisher projection) and the time term toward the time-score projection of p_t^σ . These differ unless p_t^σ is in the family, so large λ trades one for the other. This is the only bias of the moment energy with exact inputs.

4.3 What was implemented (B1–B8)

B1

`interp_time_terms` is off by default and exposed as `--interp_time_terms`. $\hat{K}_I$ is no longer saved; if on, only the r -vector diagnostics `diag_c.I`, `diag_m.I`, `diag_tau.mean.I` are.

B2

Mode `moment` (`--reg_mode`; default `moment` with `--reg_solver thomas`, `legacy` with `dense`; the config name gets `_moment`). $\hat{\Sigma}_w$ is taken at the moment-matched walkers x_{k+1} ; $\hat{m}$ is the pathwise `dt_phi.I.k`, already computed for the predictor. There is no τ anywhere. Old systems: $\hat{\Sigma} = \hat{K}_w - \hat{m} \hat{m}^\top$ with $\hat{m} = \text{barphi.p}$, and $\hat{m}$ from a 3-point non-uniform finite difference of `barphi.e` (row j at t_{j+1}) (`resolve_theta_reg.py --mode moment --results_root`). Caveat: there $\hat{K}_w$ is float32 and at y_k .

B3

$\hat{\Sigma}_w$ accumulated centred in float64. It is stored float32, like $\hat{M}$, so the offline re-solve reproduces the in-run one exactly (F3).

B4

Quadrature weights by default in the moment mode (table above); the noise-model interpretation above; the numerical check in Table 5 (holds at $\sigma = 2$).

B5

Dead-statistic masking in the Thomas solve: identity row and column, right-hand side 0 (`live` argument; unit test T3). New runs save the per-step mask. Old runs: `--mask theta` rebuilds it from the saved θ_t , which the corrector set to exactly 0 (F7); `--mask diag` is also available. `resolve_lamsweep.sh`: default `RIDGE=1e-4`, comment corrected.

B6

$w_D = 2\dot{\alpha} \cos \alpha / (d \sin \alpha)$ (T5). The Guth schedule is `--reg_schedule guth` of the moment mode; the τ -based `fixed/guth` modes of 29 Sep are removed.

B7

`forward_regularised` docstring and assert message corrected; the walker τ is computed only in legacy mode.

B8

`codes/lam_select_cv.py`, via `resolve_theta_reg.py --select (SELECT=1)`. It computes a held-out moment-matching loss (fit on even nodes, score the data term on odd nodes; equal up to a constant to $\frac{1}{2} \sum h_k \|\hat{M}_k \Theta_k - \hat{b}_k\|_{\hat{M}_k^{-1}}^2$, the linearised moment mismatch), an amplitude veto ($\max_k |\Theta_{k,i}| > 10A_i$, with A_i from block medians of $\Theta(0)$), and $\chi^2(\lambda)$. The whole path $\Theta(\lambda)$ is saved.

5 Audit findings that stand

5.1 The walkers are independent of Z

`run_experiment` passes neither `x_0` nor `x_k` to SDE, so they are separate `torch.randn` draws (`codes/sde_routines.py`, `init_interpolants_and_workers`) [verified]. With $Z \perp x$, $\mathbb{E}[\tau|x] = \dot{\alpha} d \cot \alpha$ is a constant, so $\text{Cov}_w(\phi, \tau) = 0$ and $\mathbb{E}_w[\phi\tau] = \dot{\alpha} d \cot \alpha m_w$ (sign-free) [verified] (derivation; `verify_claims.py` C2: $\mathbb{E}_w \tau = 77.28/25.12/8.12$ vs. $77.35/25.13/8.17$). Starting the walkers at Z (the bimodal port) does not help (C3 of `verify_claims.py`) [verified].

5.2 τ on true pairs, and the interpolant path

On the interpolant, $\mathbb{E}_I[\tau] = 0$ and $\mathbb{E}_I[\phi\tau] = -\dot{m}$ (sign-free), so $-\hat{c}_I$ is a score-function estimate of $\dot{m}$. It is 3.5–21 $\times$ noisier than the pathwise $\hat{m}$ ($d = 16$, `verify_claims.py` C1) [verified]. The interpolant TSM mixes two laws (A5).

5.3 Other

The ridge equivalence $(\hat{M} + \varepsilon \text{diag } \hat{M})^{-1}$ is exact (F7 reproduces θ_t to $3 \cdot 10^{-7}$) [verified]. Legacy weights are unweighted sums, so on non-uniform grids the node density weights the energy (derivation [verified]); the moment mode uses quadrature. Centring a float32 $\hat{K}$ loses precision (toy: 7% error on a variance at mean/std = 1000) [verified], which is why B3 accumulates centred.

6 Tests and validation

6.1 Unit and end-to-end tests

All run locally and pass [verified]:

- `test_moment_mode.py`:
 - T1: legacy is bit-identical to 4baf4c2 (12 random systems).
 - T2: moment mode, CODE sign, noise-free Gaussian ($\phi = (x^2, x^4)$). Max relative error $2.5 \cdot 10^{-4}$ ($N = 100$) and $1.5 \cdot 10^{-5}$ ($N = 400$) for all $\lambda \in [0, 10^4]$ and both schedules. The flipped sign gives an error of 2.17.
 - T3: masking. T4: finite difference exact on quadratics. T5: $\lambda_{\text{eff}} = \sin^2 / (\pi^2 d)$.
- `test_forward_regularised.py` (64 walkers, $M = 64$, $n_t = 20$, in-process; not `run_SDE.py`):
 - F1: legacy outputs and saved system are bit-identical to 4baf4c2.

- F2: the moment mode leaves the SDE evolution bit-identical, and the saved $\hat{m}$ equals `compute_rhs_dt_phi_I.t` exactly.
- F3: offline re-solve = in-run. F4: moment rebuilt on a legacy system (1.1% from F3). F5: `--select`. F6: `n_subsample=2`.
- F7: legacy, $\lambda = 0$, `--mask theta` reproduces θ_t .

6.2 C1, C2: 1-D ground truth

`mgd1d.py`, `run_validation.py`.

- **Ground truth** (MGD sign). Self-consistent Fokker–Planck solution of eq. (22) from $p_0 = \mathcal{N}(0, 1)$: Scharfetter–Gummel fluxes, implicit Euler with $\Delta t \leq 10^{-4}$, and the constraint re-imposed by an exponential tilt after each step (a projection; the exact dynamics conserves it). θ_t from eq. (15); $\theta^{\text{ME}}(m_t)$ by Newton.
- **Convergence** [verified]. Relative change of θ_t from ($N = 1401$, $\Delta t = 10^{-4}$) to ($N = 2801$, $\Delta t = 2.5 \cdot 10^{-5}$): $8 \cdot 10^{-5}$ ($\sigma = 10$) and 10^{-4} ($\sigma = 2$). Without the tilt: $5 \cdot 10^{-2}$.
- **Particles**. The code’s predictor–corrector in NumPy float64 (CODE sign, $\varepsilon = 10^{-6}$), with walkers *independent* of Z . The systems are solved by the real `_solve_regularised_thomas`, and λ is selected by `lam_path`.

σ	n	estimator	vs θ_t			vs θ^{ME}		
			$\lambda=0$	oracle (λ)	selected (λ)	$\lambda=0$	oracle (λ)	selected (λ)
10	1000	legacy	0.5027	0.4531 (1e-3)	0.4535 (1e-4)	0.5022	0.4525 (1e-3)	0.4529 (1e-4)
10	1000	moment, uniform	0.5027	0.3848 (1e1)	0.4510 (1e-2)	0.5022	0.3846 (1e1)	0.4504 (1e-2)
10	1000	moment, Guth	0.5027	0.1288 (1e2)	0.4481 (1e-1)	0.5022	0.1283 (1e2)	0.4475 (1e-1)
10	4000	legacy	0.4310	0.4178 (1e-4)	0.4182 (1e-5)	0.4304	0.4172 (1e-4)	0.4177 (1e-5)
10	4000	moment, uniform	0.4310	0.3458 (1e0)	0.4174 (1e-3)	0.4304	0.3453 (1e0)	0.4168 (1e-3)
10	4000	moment, Guth	0.4310	0.0897 (1e2)	0.4127 (1e-1)	0.4304	0.0898 (1e2)	0.4121 (1e-1)
10	16000	legacy	0.4116	0.4081 (1e-4)	0.4086 (1e-6)	0.4110	0.4075 (1e-4)	0.4080 (1e-6)
10	16000	moment, uniform	0.4116	0.3325 (1e0)	0.4078 (1e-3)	0.4110	0.3320 (1e0)	0.4072 (1e-3)
10	16000	moment, Guth	0.4116	0.0841 (1e2)	0.4075 (1e-2)	0.4110	0.0844 (1e2)	0.4069 (1e-2)
2	1000	legacy	1.2970	0.1504 (1e-3)	0.1504 (1e-3)	1.3350	0.1478 (1e-3)	0.1478 (1e-3)
2	1000	moment, uniform	1.2970	0.0894 (1e2)	0.0951 (1e0)	1.3350	0.0859 (1e1)	0.0868 (1e0)
2	1000	moment, Guth	1.2970	0.1045 (1e0)	0.1176 (1e1)	1.3350	0.0957 (1e0)	0.1136 (1e1)
2	4000	legacy	0.6522	0.0706 (1e-3)	0.0706 (1e-3)	0.6733	0.0850 (1e-3)	0.0850 (1e-3)
2	4000	moment, uniform	0.6522	0.0272 (1e-1)	0.0272 (1e-1)	0.6733	0.0519 (1e-1)	0.0519 (1e-1)
2	4000	moment, Guth	0.6522	0.0342 (1e0)	0.1119 (1e1)	0.6733	0.0566 (1e0)	0.1253 (1e1)
2	16000	legacy	0.3385	0.0562 (1e-3)	0.0648 (1e-4)	0.3524	0.0770 (1e-3)	0.0842 (1e-4)
2	16000	moment, uniform	0.3385	0.0438 (1e-1)	0.0455 (1e-2)	0.3524	0.0682 (1e-1)	0.0693 (1e-2)
2	16000	moment, Guth	0.3385	0.0498 (1e-1)	0.0503 (1e0)	0.3524	0.0724 (1e-1)	0.0739 (1e0)

Table 3: C1, bimodal. Relative error $\|\Theta - \vartheta\|_G / \|\vartheta\|_G$ (Fisher metric of p_t^σ , $0.05 \leq t \leq 0.95$, MGD sign) against MGD’s θ_t and against $\theta^{\text{ME}}(m_t)$. $\lambda=0$ is the per-step $\hat{\theta}$. Oracle: best λ of the grid; selected: `lam_path` (held-out loss + veto). $n_t = 2000$, one seed.

Reading (one seed per row; all numbers from `results.json`, full λ curves there) [verified].

- $\sigma = 2$ is the clean regime: $h\sigma^2 = 0.002$, and the B4 noise model holds (Table 5).
 - The per-step error falls like $n^{-1/2}$: C2 0.71/0.35/0.18; C1 1.30/0.65/0.34 vs. θ_t .
 - *Legacy* is U-shaped in λ . Small λ removes noise, but at large λ the error climbs back up, to 0.68–0.70 (C2) and 0.58–0.59 (C1) at $\lambda = 10^2$: it converges to the wrong target.
 - *Moment, uniform* flattens instead. C2 error at $\lambda = 10^2$: 0.065/0.018/0.019, against a best of 0.065/0.017/0.014. The flatness is to within 0.005 over $\lambda \in [10^{-1}, 10^2]$; it is not exactly flat. It is the best oracle in every $\sigma = 2$ row except C2, $n = 1000$ (Guth schedule, 0.064 vs. 0.065), and CV selection lands on or next to the oracle (C2 selected = oracle; C1 0.095/0.027/0.046 vs. oracle 0.089/0.027/0.044).

σ	n	estimator	vs $\theta_t = \theta^{\text{ME}}$		
			$\lambda=0$	oracle (λ)	selected (λ)
10	1000	legacy	0.2272	0.1989 (1e-4)	0.2000 (1e-5)
10	1000	moment, uniform	0.2272	0.1553 (1e2)	0.1967 (1e-2)
10	1000	moment, Guth	0.2272	0.0985 (1e2)	0.1941 (1e-1)
10	4000	legacy	0.1768	0.1688 (1e-5)	0.1688 (1e-5)
10	4000	moment, uniform	0.1768	0.1283 (1e2)	0.1681 (1e-3)
10	4000	moment, Guth	0.1768	0.0519 (1e2)	0.1641 (1e-1)
10	16000	legacy	0.1525	0.1501 (1e-5)	0.1505 (1e-6)
10	16000	moment, uniform	0.1525	0.1147 (1e1)	0.1484 (1e-2)
10	16000	moment, Guth	0.1525	0.0376 (1e2)	0.1495 (1e-2)
2	1000	legacy	0.7098	0.0905 (1e-3)	0.0905 (1e-3)
2	1000	moment, uniform	0.7098	0.0649 (1e2)	0.0649 (1e2)
2	1000	moment, Guth	0.7098	0.0641 (1e1)	0.0641 (1e1)
2	4000	legacy	0.3474	0.0276 (1e-3)	0.0276 (1e-3)
2	4000	moment, uniform	0.3474	0.0173 (1e0)	0.0173 (1e0)
2	4000	moment, Guth	0.3474	0.0177 (1e0)	0.0250 (1e1)
2	16000	legacy	0.1813	0.0180 (1e-3)	0.0249 (1e-4)
2	16000	moment, uniform	0.1813	0.0141 (1e-1)	0.0141 (1e-1)
2	16000	moment, Guth	0.1813	0.0156 (1e0)	0.0156 (1e0)

Table 4: C2, Gaussian ($\phi = x^2$): all projections coincide, $\theta_t = \theta^{\text{ME}} = 1/(2v_t)$. Same conventions as Table 3.

- In C1 the moment error stops decreasing between $n = 4000$ and 16000 ($0.027 \rightarrow 0.044$, one seed). This is consistent with a floor from the projection mismatch of A4 (θ_t and θ^{ME} differ by 3.6% at $\sigma = 2$) [hypothesis].
- *Moment, Guth schedule*: similar at its best λ , but the error rises again for $\lambda \geq 10$ (at $\lambda = 10^2$: C1 0.26–0.59, C2 up to 0.14). Its data weight $w_D \rightarrow 0$ as $t \rightarrow 1$, so the time term alone carries the end. CV selection is also worse for it (C1, $n = 4000$: 0.112 selected vs. 0.034 oracle). This supports keeping it as a comparison option only.
- $\sigma = 10$ **with** $n_t = 2000$ **is under-resolved**: $h\sigma^2 = 0.05$. B4 shows a deterministic bias of $\hat{\theta}_k$: mean z grows like $\sqrt{n}$, and it falls faster than $\sqrt{h}$ from $n_t = 1000$ to 4000 , i.e. the bias is $O(h)$. So all estimators that trust the data term sit at a floor: 0.41–0.45 (C1, selected) and 0.15–0.20 (C2). The held-out loss scores against the same biased data, so it cannot see this. Only large- λ Guth solutions, which rely mostly on the time term, get below it (oracle 0.08–0.13 in C1, 0.04–0.10 in C2). A resolved $\sigma = 10$ test needs $n_t \gtrsim 2 \cdot 10^4$ (hours on this CPU); not run. For reference, the bimodal experiment uses $n_t = 5 \cdot 10^4$ ($h\sigma^2 = 2 \cdot 10^{-3}$).

6.3 B4: per-step noise model

[verified]: at $\sigma = 2$ the model $\text{Cov}(\hat{\theta}_k) = 2\hat{M}_k^{-1}/(nh\sigma^2)$, white, holds: $\mathbb{E}[z^2] \in [0.85, 1.06]$ and $|\text{lag-1}| \leq 0.05$ for every n and n_t tested. At $\sigma = 10$, $n_t \leq 4000$, a smooth $O(h)$ bias dominates (lag-1 up to 0.99); the noise part cannot be isolated there.

6.4 C3: existing turbulence systems (Jean Zay; not run)

After pushing this code, on Jean Zay (from turbulence/launch; P selects one zfloor system):

```
P="*zfloor_two_phase_reg1e-4*seed_900*.pt"
SYSTEM_PATTERN="$P" MODE=legacy MASK=none RIDGE=1e-4 LAMS="0" bash resolve_lamsweep.sh
SYSTEM_PATTERN="$P" MODE=legacy MASK=theta RIDGE=1e-4 LAMS="0" bash resolve_lamsweep.sh
SYSTEM_PATTERN="$P" MODE=moment MASK=theta RIDGE=1e-4 SELECT=1 \
  LAMS="0 1e-8 1e-7 1e-6 1e-5 1e-4 1e-3 1e-2 1e-1 1" bash resolve_lamsweep.sh
# then, from the project root, for that config:
python notes/guth_reg_audit_0930/c3_check.py <config> --root turbulence --ridge 1e-4
```

c3_check.py prints: (a) masked $\Theta(0)$ vs. θ_t ; (b) the B5 [hypothesis]: are the exploded entries of the unmasked $\Theta(0)$ the ones with $\theta_t = 0$?; (c) the moment path (selected λ , vetoes, χ^2) and the distance to the block averages of θ_t .

case	σ	n	n_t	$\mathbb{E}[z^2]$ per component	lag-1 autocorr.
C1	10	1000	2000	1.04, 12.54, 1.29, 19.78	+0.184, +0.915, +0.424, +0.956
C1	10	4000	2000	2.46, 37.50, 3.78, 59.06	+0.592, +0.972, +0.767, +0.986
C1	10	16000	2000	5.01, 133.59, 10.30, 229.89	+0.737, +0.990, +0.930, +0.995
C1	10	4000	1000	19.34, 395.45, 30.31, 574.19	+0.933, +0.993, +0.939, +0.993
C1	10	4000	4000	1.09, 4.53, 1.08, 6.72	+0.137, +0.715, +0.311, +0.844
C1	2	1000	2000	1.04, 1.04, 1.00, 0.95	+0.001, +0.021, +0.033, -0.014
C1	2	4000	2000	0.98, 0.94, 1.00, 0.97	+0.017, -0.034, +0.050, +0.051
C1	2	16000	2000	1.06, 1.02, 1.04, 1.03	+0.002, -0.027, -0.022, -0.002
C1	2	4000	1000	0.99, 0.96, 0.99, 0.85	-0.032, -0.035, +0.020, -0.032
C1	2	4000	4000	0.98, 1.02, 1.01, 0.97	-0.004, -0.013, +0.025, -0.022
C2	10	1000	2000	3.36	+0.640
C2	10	4000	2000	7.96	+0.818
C2	10	16000	2000	23.43	+0.930
C2	2	1000	2000	1.01	+0.018
C2	2	4000	2000	0.95	-0.040
C2	2	16000	2000	1.01	-0.036

Table 5: $z_k = \sqrt{nh\sigma^2/2} L_k^\top (\hat{\theta}_k - \theta_t)$, $L_k L_k^\top = \hat{M}_k + \varepsilon D_k$, $0.05 \leq t \leq 0.95$ (MGD sign; reference θ_t from Fokker–Planck in C1, exact in C2). The model predicts $\mathbb{E}[z^2] = 1$ and lag-1 autocorrelation 0. A deterministic $O(h)$ bias of $\hat{\theta}_k$ adds to $\mathbb{E}[z^2]$ in proportion to nh .

7 Open

- C3 on Jean Zay (above); C4 (one smoke run of `run_SDE.py` saving the moment system) only after that, and only on request.
- Bimodal port (`bimodal_experiment/sde_routines_scalar_reg.py`, `codes/lam_selection.py`): still legacy. Porting the moment mode there and re-running the REG figure is the A8 follow-up.
- Sizes: a moment system has the same size as a legacy one ($\hat{M}$ and $\hat{\Sigma}_w$ float32, plus r -vectors): 37.9 GB/seed at $n_t = 60\,000$, $r = 281$ [reported, not re-run] (the legacy size from 29 Sep).
- The draft’s eqs. (4), (6), (8) and the MGD eq. numbers are taken from the external review [derived, external review].

A Scripts (notes/guth_reg_audit_0930/)

- `verify_claims.py`, `verify_bias.py`: the checks of the first version of this note (`verify_bias.py` computes θ^{ME} , see A5).
- `test_moment_mode.py`, `test_forward_regularised.py`: tests T1–T5, F1–F7.
- `mgd1d.py`, `run_validation.py`, `make_tables.py`: C1, C2, B4 (`results.json`, `validation_log.txt`).
- `c3_check.py`: C3, to run on Jean Zay.